

B = 64

T = 256

C = 384

Model learns:

$$P(x_{t+1} | x_0, x_1, \dots, x_t)$$

input: "hello"

↓ character level tokenisation

$$W_Q, W_K, W_V$$

$$Q = X W_Q$$

$$K = X W_K$$

$$V = X W_V$$

$$Q, K, V \in \mathbb{R}^{T \times 384}$$

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

(T × 384) (T × T) (T × 384)

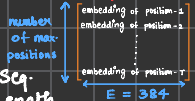
Randomly initialised

$$W_E \leftarrow W_E - \alpha \frac{\partial L}{\partial W_E}$$

$$W_E \in \mathbb{R}^{49 \times 384}$$



$$W_P \in \mathbb{R}^{256 \times 384}$$



h

e

l

l

o

token IDs

Vocab size = V

11

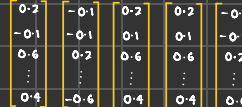
4

19

19

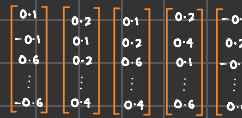
27

embedding lookup



adding positional encoding

$$X \in \mathbb{R}^{T \times 384}$$



for token x_t at position t

To transformer Blocks

$$h_t = E[x_t] + P[t]$$

Embedding

Positional Embedding

$$q_1^k, k_1^k, v_1^k, q_2^k, k_2^k, v_2^k, \dots, q_T^k, k_T^k, v_T^k$$

Attention



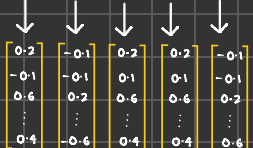
Convert to logits

$$\text{logits}_t = h_t^T W_{lm} + b$$

V

584

T = 256



h_1

h_2

h_3

...

h_t

only this needs to be softmaxed during inference.

(x depth) = 6 for shakepave

Seq. length

$$QK^T$$

t+1

t+1

$$q_{t+1}^k, \dots, q_{t+1}^k, k_{t+1}^k$$

384

t+1

384

$$Q$$

$$K^T$$

$$V$$

only extra col.

Only this row need to calculate

→ So we cache KV after prefill

→ No need to store Q.

→ With each extra token, K and V increase by a row.

will compute avg. crossentropy while training.